

Spatial Similarity Analysis Report: Land Cover & PM 2.5 Correlation

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1 Objective

To calculate the weighted Bray-Curtis similarity score between three geographic locations (WomenDGP, IITG, and Jadavpur University) based on compositional land cover data, adjusted for their predictive impact on localized PM 2.5 variations.

2 Input Data & Environmental Weights

The analysis utilizes the following land cover percentages extracted from the classified datasets, alongside their assigned environmental weights (w_i).

Land Cover Type	Weight (w_i)	WomenDGP (%)	IITG (%)	JU (%)
Road & Infrastructure	0.70	2.73	27.87	41.28
Greenery	0.15	81.39	35.28	44.03
Empty Land	0.10	15.68	12.34	11.99
Water	0.05	0.20	24.52	2.69

Note: Because "Road" and "Infrastructure" were combined in the raw data, their individual literature-derived weights (0.35 each) were summed to 0.70 to maintain mathematical integrity.

3 Methodology

The calculations utilize the Corrected Weighted Bray-Curtis formula to find the spatial similarity between locations A and B:

$$S_{final} = 1 - \frac{\sum_{i=1}^n w_i |A_i - B_i|}{\sum_{i=1}^n w_i (A_i + B_i)} = 1 - \frac{N}{D} \quad (1)$$

Where N represents the weighted absolute differences (dissimilarity magnitude) and D represents the weighted sums (total potential magnitude).

4 Calculations & Final Results

1. IITG vs. Jadavpur University (JU)

- **Similarity Score: 0.8155**
- **Interpretation (Highly Similar):** Both locations feature significant built-up areas and moderate greenery. Because the "Road & Infrastructure" category carries the heaviest predictive weight (0.70), the mathematical model recognizes that both locations will actively generate and trap vehicular/industrial particulates in a very similar manner.

2. WomenDGP vs. IITG

- **Similarity Score: 0.3933**
- **Interpretation (Highly Dissimilar):** WomenDGP is completely dominated by Greenery (81.39%) with almost non-existent Road & Infrastructure (2.73%), contrasting sharply with IITG's highly developed and balanced profile.

3. WomenDGP vs. Jadavpur University (JU)

- **Similarity Score: 0.3702**
- **Interpretation (Highly Dissimilar):** The severe lack of primary PM 2.5 emission sources in WomenDGP makes its air quality dynamics entirely different from the highly urbanized JU campus.

5 Conclusion

If comparing the actual PM 2.5 readings for these three locations, **IITG and JU** are expected to share highly similar pollution trends and magnitude spikes (Similarity: 81.55%). Conversely, **WomenDGP** serves as an extreme ecological outlier. Due to its massive natural filtering capacity and lack of emission surfaces, it should consistently record drastically lower, more stable AQI levels compared to the other two campuses.